A-A-2579
May 2, 1988

COMMERCIAL ITEM DESCRIPTION

PSYCHROMETER, SLING, POCKET TYPE

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification GG-P-725B, Type V.

This description covers a psychrometer to be used for determining relative humidity of atmospheric air by means of wet- and dry-bulb temperature readings.

Salient charcteristics.

Materials.
Corrosion-resistant metal wherever possible. Any metal causing galvanic action shall be avoided.
Thermometer stem shall be standard glass.

Design and construction.
General requirements. Shall consist of two identical mercurial thermometers, mounted parallel, one fitted with a wick for wetting the bulb and other components consistent with the type described herein. See Figures 1 and 2.

Thermometer. Shall consist of a stem, expansion space, bulb, mercury indicator and a button top.

Thermometer range and graduation.
The temperature range shall be 30 to 110 degrees F. The scale graduations shall be etched or permanent stain imprinted in the glass in whole degrees. The graduation lines shall be clear cut, straight and perpendicular to the axis of the stem. The minor graduation lines shall lie to the left of the axis and shall terminate in line with the axis. The 5 and 10 degree lines extend across the axis to the right. The graduations shall be numbered at each multiple of 10 degrees.

Handle and sling. Shall be provided with a 3-3/4 inch long by 5/16 inch in diameter, partly knurled corrosion resistant metal handle which shall be attached to a 4 inch section of corrosion resistant metal chain. This shall be attached to the thermometer frame.

Length. The total length of the bulb and stem shall be 4-3/4 + 1/8 inch. Minimum length of graduated portion shall be 2-1/4 inch.

Pigment. All etched graduations and numerals shall be filled with a black pigment.

FSC 6685

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:24Z
Check the source to verify that this is the current version before use.

Stain. Permanent stain imprinting on the glass stem is permissable. Imprinted data shall be sharp and clear.

Wick. The wick shall be chemically clean muslin or bleached cotton sewn or woven into a tubular shape. The lower end of the shall be appropriately secured immediately below the wet-bulb thermometer.

Pocket case. Each psychrometer shall be furnished with a pocket case, fitted with a protective metal insert, and designed to protect the complete instrument, including the handle and chain.

Calculator. Shall be provided with a plastic calculator for determining the relative humidity.

Instrument container. When specified, each psychrometer, including pocket case and a plastic calculator, shall be furnished in a wood or plastic case.

Workmanship. Each pyschrometer shall be free from defects which impair their durability, appearance or serviceability.

Test procedures.

Thermometer accuracy. A standard test thermometer of which the calibration is known to be within 0.1 degree F. over the operating range of thermometers to be tested shall be used as referenoes for the accuracy testing. The reference thermometer shall be placed centrally among the thermometers being tested and immersed to top of the liquid column in a liquid bath being constantly agitated. Both the reference and test thermometers shall have their bulbs placed in approximately the same level in the bath. The temperature of the liquid bath shall be held constant to ± 0.1 degree F. of the test point selected. Data shall not be recorded until the temperature stabilizes.

The thermometers to be tested shall be compared with the reference thermometer in an ice bath and at a series of points taken over the entire range. Test points shall be selected such that the uppermost and lowermost points shall fall within 0.5 degree F. of the end points of the specified ranges and the interval between consecutive test points shall not be closer than 10 scale divisions and no further apart than 50 scale divisions. An ice point should always be included as one of the test points, where an ice point is covered by the range of the thermometer. The thermometer shall indicate true temperature within the tolerance of ± 0.5 degree F. The change in error for an interval of 20 degree F. shall be no greater than 0.5 degree F. on any part of the scale. The change in the ice point during a 90 day period shall be not more than ± 0.1 degree F. Each pair of thermometers furnished with a psychrometer shall be in in agreement within 0.5 degree F. throughout their range. The maximum allowable error shall be ± 1 degree F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:24Z
Check the source to verify that this is the current version before use.

Expansion space. The thermometer shall be immersed completely for five minutes in a water bath having a temperature of 130 ± 0.5 degree F. The thermometer shall then be cooled in air to ambient temperature. After this test, the thermometer shall indicate true temperatures within ± 0.5 degrees F.

Handle/sling test. A one pound weight shall be attached to the sling in place of the psychrometer. The weight shall be rotated about the handle at a rate of approximately 3 revolutions per second for ten seconds. There shall be no damage to the handle or sling.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein.

Sampling. Sampling for inspection and tests shall be performed in accordance with MIL-STD-105. For purposes of sampling, a lot shall consist of all psychrometers of the same description manufactured under the same conditions at essentially the same time.

Sampling for visual and dimensional inspection. A random sample shall be taken from each lot in accordance with Inspection Level I. The AQL shall not exceed 1.5 percent defective. The sample shall be one psychrometer for visual and dimensional inspection.

Sampling for testing. A random sample shall be taken from each lot in accordance with Inspection Level S-2. The AQL shall not exceed 1.0 defective. The sample shall be one unit.

Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, The Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing supplier's employees to undue hazards from the recovered materials.

Preservation, packaging, packing, labeling and marking. The preservation, packaging, packing, labeling and marking shall be as specified in the contract or purchase order.

Note: Purchaser shall specify NSN 6685-00-826-1662 when ordering.

MILITARY INTERESTS:

NONE

PREPARING ACTIVITY:

GSA - FSS

Source: https://assist.dla.mil -- Downloaded: 202610-26T08:24Z
Check the source to verify that this is the current version before use.

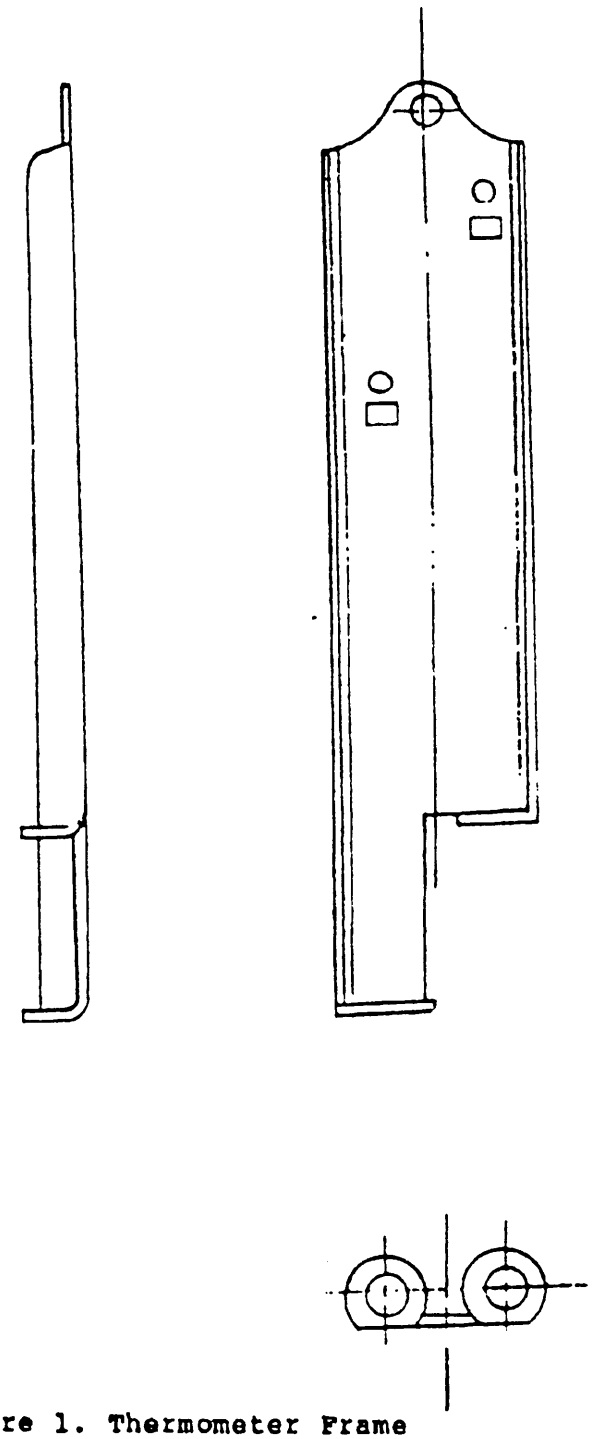


Figure 1. Thermometer Frame

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:24Z
Check the source to verify that this is the current version before use.

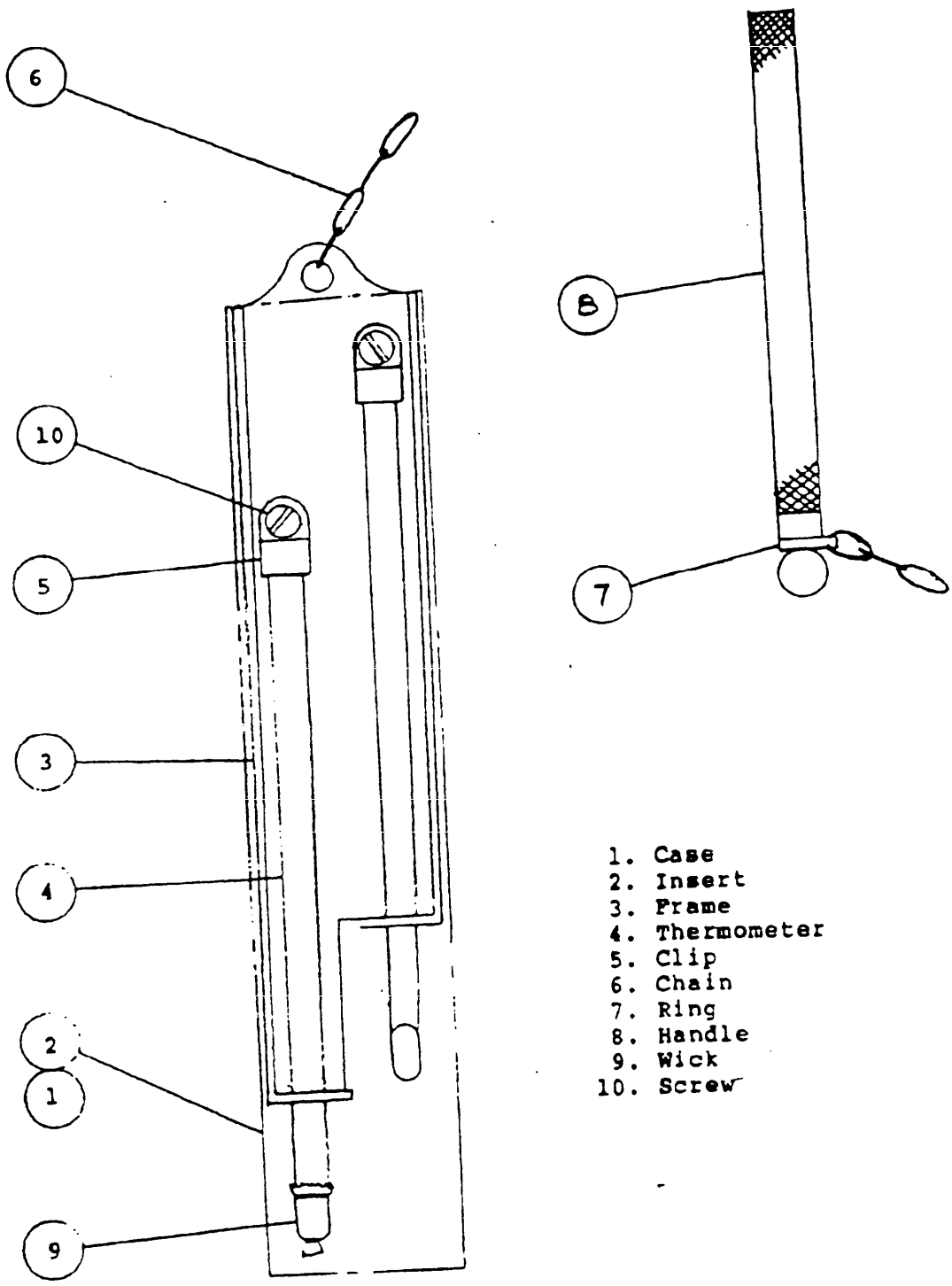


Figure 2. Psychrometer

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:24Z
Check the source to verify that this is the current version before use.